МИНИСТЕРСТВО НАУКИ И ВЫСШЕГО ОБРАЗОВАНИЯ РФ

ФЕДЕРАЛЬНОЕ ГОСУДАРСТВЕННОЕ БЮДЖЕТНОЕ ОБРАЗОВАТЕЛЬНОЕ УЧРЕЖДЕНИЕ ВЫСШЕГО ОБРАЗОВАНИЯ «ЛИПЕЦКИЙ ГОСУДАРСТВЕННЫЙ ТЕХНИЧЕСКИЙ УНИВЕРСИТЕТ»

Институт компьютерных наук
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Лабораторная работа №6

по дисциплине «Теория обработки больших массивов данных»

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1. Задание кафедры

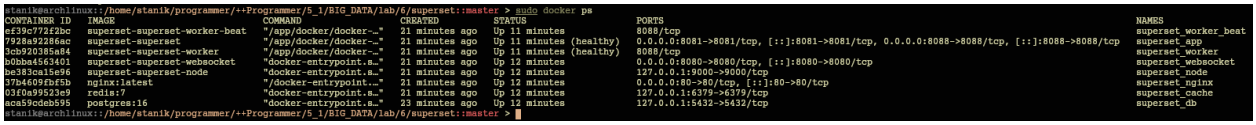
Запустите Superset, используя Docker и выведите дэшборды (WebUI).

2. Ход работы

** В ходе выполнения работы использовалось ядро GNU/Linux v6.15.4, Docker v28.3.0, Superset v5.0.0 **

Клонировал репозиторий Superset (https://github.com/apache/superset) docker-compose.yml был уже настроен и содержал все необходимые зависимости для корректной работы. В том числе готовые базы данных.

Все контейнеры были запущены иуспешно функционируют (UP 11 minutes status)



Для того, чтобы запустить эти контейнеры пришлось сначала:
sudo docker compose up db — запустить контейнер с БД (иначе краш из-за того что контейнер superset-init будет требовать базы данных из examples, которых еще нет. Чтобы это исправить автоматически, можно было добавить «depends-on» для контейнера superset-init в docker-compose.yml, потому что из-за мягкого ограничивающего условия на моей системе все равно DB не успевал встать).

```
    depends_on:
      db:
        condition: service_started # Это условие мешало полноценной
инициализации DB
```

Когда я разобрался с одной проблемой, получил вторую.

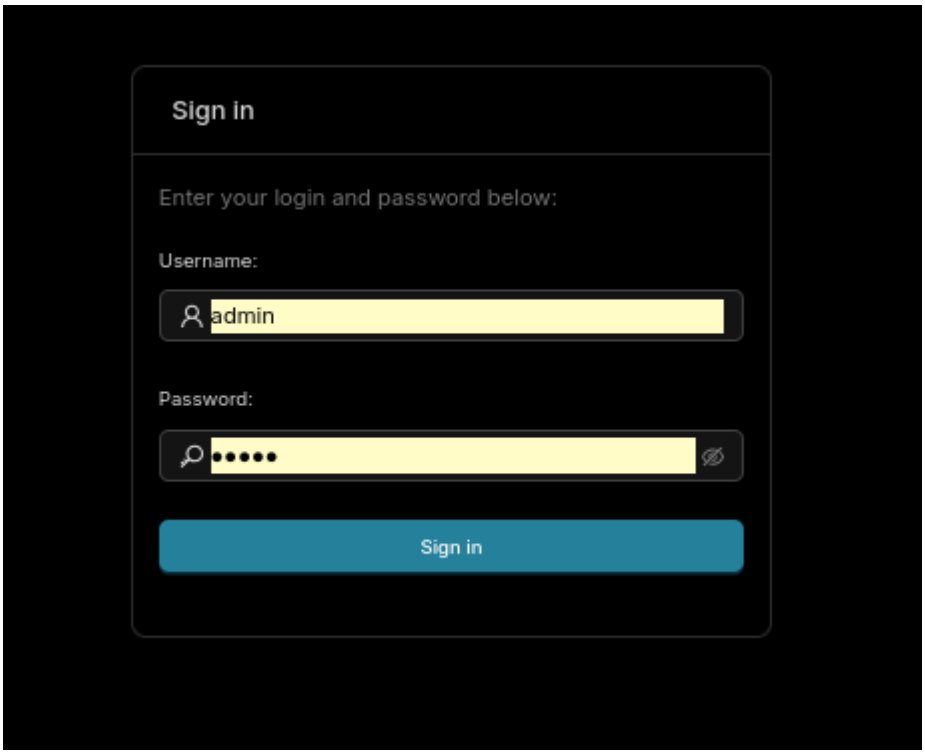
Вторая проблема была с загрузкой webpack — он просто не мог

загрузить зависимости

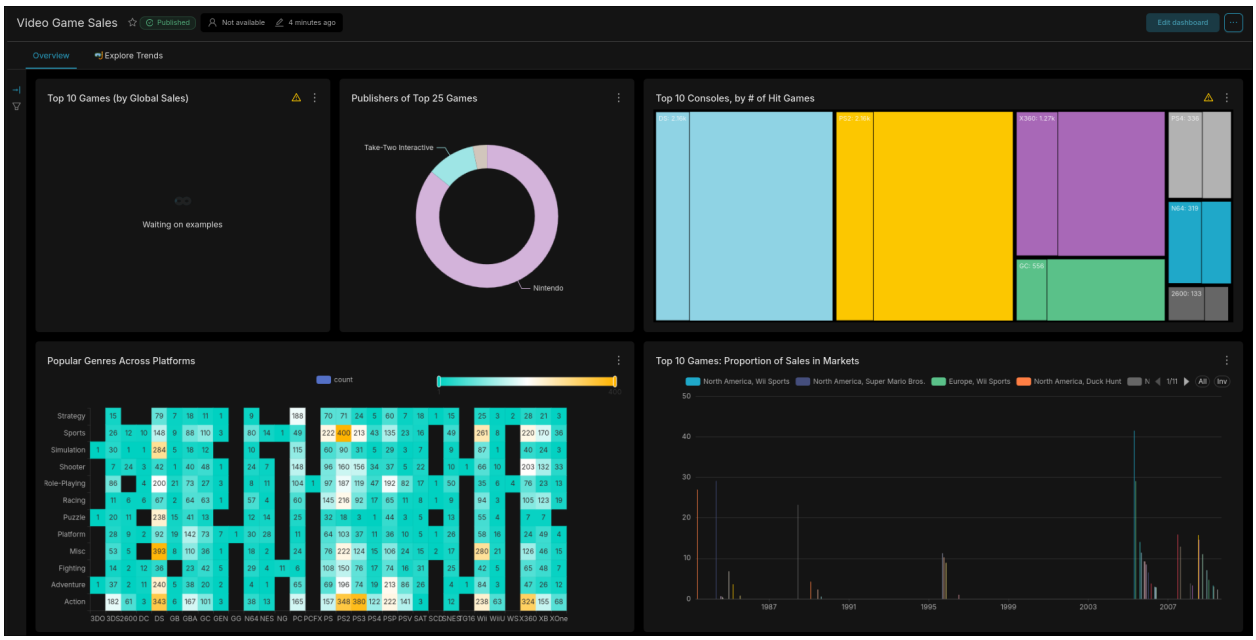
```
superset_node          | <s> [webpack.Progress] 35% building 7/15 entries 254/288 dependencies 83/100 modules
superset_node          | <s> [webpack.Progress] 39% building 8/15 entries 375/436 dependencies 94/145 modules
superset_node          | <s> [webpack.Progress] 39% building 8/15 entries 826/900 dependencies 138/387 modules
superset_node          | <s> [webpack.Progress] 39% building 8/15 entries 1400/1489 dependencies 225/482 modules
superset_node          | <s> [webpack.Progress] 39% building 8/15 entries 1798/1900 dependencies 292/672 modules
superset_node          | <s> [webpack.Progress] 39% building 8/15 entries 2400/2598 dependencies 332/846 modules
superset_node          | <s> [webpack.Progress] 39% building 8/15 entries 3174/3200 dependencies 365/1067 modules
superset_node          | node:events:502
superset_node          |       throw er; // Unhandled 'error' event
superset_node          |       ^
superset_node          |
superset_node          | Error: read ECONNRESET
superset_node          |     at Pipe.onStreamRead (node:internal/stream_base_commons:218:20)
superset_node          | Emitted 'error' event on Socket instance at:
superset_node          |     at emitErrorNT (node:internal/streams/destroy:169:8)
superset_node          |     at emitErrorCloseNT (node:internal/streams/destroy:128:3)
superset_node          |     at process.processTicksAndRejections (node:internal/process/task_queues:82:21) {
superset_node          |   errno: -104,
superset_node          |   code: 'ECONNRESET',
superset_node          |   syscall: 'read'
superset_node          | }
superset_node          |
superset_node          | Node.js v20.19.6
superset_node exited with code 1
```

Чтобы это исправить достаточно было смены IP адреса системы, на которой запускался контейнер, на адрес из другого региона.

И, наконец, теперь localhost:8088 обновился: вместо бесконечно крутящегося логотипа Superset, меня встретил экран авторизации пользователя.



Дешборд Video Game Sales:



Чарт Unicode Cloud:



3. Вывод

В ходе выполнения лабораторной работы было запущен комлекс контейнеров для работы Superset.

Приложение 1. docker-compose.yml

#

```
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# distributed under the License is distributed on an "AS IS" BASIS,
# WITHOUT WARRANTIES OR CONDITIONS OF ANY KIND, either express or implied.
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# limitations under the License.
#


# -----------------------------------------------------------------------
# We don't support docker compose for production environments.
# If you choose to use this type of deployment make sure to
# create you own docker environment file (docker/.env) with your own
# unique random secure passwords and SECRET_KEY.
#
# For verbose logging during development:
# - Set SUPERSET_LOG_LEVEL=debug in docker/.env-local for detailed Superset logs
# -----------------------------------------------------------------------
x-superset-user: &superset-user root
x-superset-volumes: &superset-volumes
  # /app/pythonpath_docker will be appended to the PYTHONPATH in the final container
  - ./docker:/app/docker
  - ./superset:/app/superset
  - ./superset-core:/app/superset-core
  - ./superset-frontend:/app/superset-frontend
  - superset_home:/app/superset_home
  - ./tests:/app/tests
  - superset_data:/app/data
x-common-build: &common-build
  context: .
  target: ${SUPERSET_BUILD_TARGET:-dev} # can use `dev` (default) or `lean`
  cache_from:
    - apache/superset-cache:3.10-slim-trixie
  args:
    DEV_MODE: "true"
    INCLUDE_CHROMIUM: ${INCLUDE_CHROMIUM:-false}
    INCLUDE_FIREFOX: ${INCLUDE_FIREFOX:-false}
    BUILD_TRANSLATIONS: ${BUILD_TRANSLATIONS:-false}
    LOAD_EXAMPLES_DUCKDB: ${LOAD_EXAMPLES_DUCKDB:-true}

services:
  nginx:
    env_file:
      - path: docker/.env # default
        required: true
      - path: docker/.env-local # optional override
        required: false
    image: nginx:latest
    container_name: superset_nginx
    restart: unless-stopped
```

```
  ports:
    - "80:80"
  extra_hosts:
    - "host.docker.internal:host-gateway"
  volumes:
    - ./docker/nginx/nginx.conf:/etc/nginx/nginx.conf:ro
    - ./docker/nginx/templates:/etc/nginx/templates:ro

redis:
  image: redis:7
  container_name: superset_cache
  restart: unless-stopped
  ports:
    - "127.0.0.1:6379:6379"
  volumes:
    - redis:/data

db:
  env_file:
    - path: docker/.env # default
      required: true
    - path: docker/.env-local # optional override
      required: false
  image: postgres:16
  container_name: superset_db
  restart: unless-stopped
  ports:
    - "127.0.0.1:5432:5432"
  volumes:
    - db_home:/var/lib/postgresql/data
    - ./docker/docker-entrypoint-initdb.d:/docker-entrypoint-initdb.d

superset:
  env_file:
    - path: docker/.env # default
      required: true
    - path: docker/.env-local # optional override
      required: false
  build:
    <<: *common-build
  container_name: superset_app
  command: ["/app/docker/docker-bootstrap.sh", "app"]
  restart: unless-stopped
  ports:
    - 8088:8088
    # When in cypress-mode ->
    - 8081:8081
  extra_hosts:
    - "host.docker.internal:host-gateway"
  user: *superset-user
  depends_on:
    superset-init:
      condition: service_completed_successfully
  volumes: *superset-volumes
  environment:
    SUPERSET__SQLALCHEMY_EXAMPLES_URI: "duckdb:////app/data/examples.duckdb"
```

```
superset-websocket:
  container_name: superset_websocket
  build: ./superset-websocket
  ports:
    - 8080:8080
  extra_hosts:
    - "host.docker.internal:host-gateway"
  depends_on:
    - redis
  # Mount everything in superset-websocket into container and
  # then exclude node_modules and dist with bogus volume mount.
  # This is necessary because host and container need to have
  # their own, separate versions of these files. .dockerignore
  # does not seem to work when starting the service through
  # docker compose.
  #
  # For example, node_modules may contain libs with native bindings.
  # Those bindings need to be compiled for each OS and the container
  # OS is not necessarily the same as host OS.
  volumes:
    - ./superset-websocket:/home/superset-websocket
    - /home/superset-websocket/node_modules
    - /home/superset-websocket/dist

    # Mount config file. Create your own docker/superset-websocket/config.json
    # for custom settings, then point to it here. Do not use this example in production.
    - ./docker/superset-websocket/config.example.json:/home/superset-websocket/config.json:ro
  environment:
    - PORT=8080
    - REDIS_HOST=redis
    - REDIS_PORT=6379
    - REDIS_SSL=false

superset-init:
  build:
    <<: *common-build
  container_name: superset_init
  command: ["/app/docker/docker-init.sh"]
  env_file:
    - path: docker/.env # default
      required: true
    - path: docker/.env-local # optional override
      required: false
  depends_on:
    db:
      condition: service_started
    redis:
      condition: service_started
  user: *superset-user
  volumes: *superset-volumes
  environment:
    SUPERSET__SQLALCHEMY_EXAMPLES_URI: "duckdb:////app/data/examples.duckdb"
  healthcheck:
    disable: true
```

```
  superset-node:
    build:
      context: .
      target: superset-node
      args:
        # This prevents building the frontend bundle since we'll mount local folder
        # and build it on startup while firing docker-frontend.sh in dev mode, where
        # it'll mount and watch local files and rebuild as you update them
        DEV_MODE: "true"
        BUILD_TRANSLATIONS: ${BUILD_TRANSLATIONS:-false}
    environment:
      # set this to false if you have perf issues running the npm i; npm run dev in-docker
      # if you do so, you have to run this manually on the host, which should perform better!
      BUILD_SUPERSET_FRONTEND_IN_DOCKER: true
      NPM_RUN_PRUNE: false
      SCARF_ANALYTICS: "${SCARF_ANALYTICS:-}"
      # configuring the dev-server to use the host.docker.internal to connect to the backend
      superset: "http://superset:8088"
    ports:
      - "127.0.0.1:9000:9000"  # exposing the dynamic webpack dev server
    container_name: superset_node
    command: ["/app/docker/docker-frontend.sh"]
    env_file:
      - path: docker/.env # default
        required: true
      - path: docker/.env-local # optional override
        required: false
    volumes: *superset-volumes

  superset-worker:
    build:
      <<: *common-build
    container_name: superset_worker
    command: ["/app/docker/docker-bootstrap.sh", "worker"]
    env_file:
      - path: docker/.env # default
        required: true
      - path: docker/.env-local # optional override
        required: false
    environment:
      CELERYD_CONCURRENCY: 2
    restart: unless-stopped
    depends_on:
      superset-init:
        condition: service_completed_successfully
    user: *superset-user
    volumes: *superset-volumes
    extra_hosts:
      - "host.docker.internal:host-gateway"
    healthcheck:
        test: ["CMD-SHELL", "celery -A superset.tasks.celery_app:app inspect ping -d celery@$
$HOSTNAME"]
    # Bump memory limit if processing selenium / thumbnails on superset-worker
    # mem_limit: 2038m
    # mem_reservation: 128M

  superset-worker-beat:
```

```
    build:
      <<: *common-build
    container_name: superset_worker_beat
    command: ["/app/docker/docker-bootstrap.sh", "beat"]
    env_file:
      - path: docker/.env # default
        required: true
      - path: docker/.env-local # optional override
        required: false
    restart: unless-stopped
    depends_on:
      - superset-worker
    user: *superset-user
    volumes: *superset-volumes
    healthcheck:
      disable: true

  superset-tests-worker:
    build:
      <<: *common-build
    container_name: superset_tests_worker
    command: ["/app/docker/docker-bootstrap.sh", "worker"]
    env_file:
      - path: docker/.env # default
        required: true
      - path: docker/.env-local # optional override
        required: false
    profiles:
      - optional
    environment:
      DATABASE_HOST: localhost
      DATABASE_DB: test
      REDIS_CELERY_DB: 2
      REDIS_RESULTS_DB: 3
      REDIS_HOST: localhost
      CELERYD_CONCURRENCY: 8
    network_mode: host
    depends_on:
      superset-init:
        condition: service_completed_successfully
    user: *superset-user
    volumes: *superset-volumes
    healthcheck:
        test: ["CMD-SHELL", "celery inspect ping -A superset.tasks.celery_app:app -d celery@$
$HOSTNAME"]

volumes:
  superset_home:
    external: false
  db_home:
    external: false
  redis:
    external: false
  superset_data:
    external: false
```